

CAUCASUS UNIVERSITY · SCHOOL OF TECHNOLOGY

Python Programming II

Meeting 2 · Numbers, Floats & Functions

binary representation · Newton-Raphson · functions

Lecturer: **Prof. Maksim Iavich**

miavich@cu.edu.ge · Scientific Cyber Security Association



RECAP

From last time: bisection

Square root by halving the search space

Task

square root by bisection

Write (or trace) the Python program described by the title above.

The worked solution is on the next slide — try it yourself first.

Recap — square root by bisection

```
x = 25
epsilon = 0.01
numGuesses = 0
low = 1.0
high = x
ans = (high + low)/2.0
while abs(ans**2 - x) >= epsilon:
    numGuesses += 1
    if ans**2 < x:
        low = ans
    else:
        high = ans
    ans = (high + low)/2.0
print('numGuesses = ' + str(numGuesses))
print(str(ans) + ' is close to square root of ' + str(x))
```

Dealing with float's

- Floats approximate real numbers, but it is useful to understand how.
- **Decimal number:**
  - $302 = 3 \cdot 10^2 + 0 \cdot 10^1 + 2 \cdot 10^0$
- **Binary number:**
  - $10011 = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0$ $(= 16 + 2 + 1 = 19)$
- Internally, the computer represents numbers in binary.

Converting an integer to binary

- **Consider $x = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 10011$**
- Remainder relative to 2 (x % 2) gives the last binary bit.
- Dividing x by 2 (x // 2) shifts all bits right: x//2 = 1001.
- Keep doing successive divisions; each remainder gives the next bit, and so on.

Task

Integer → binary string

Write (or trace) the Python program described by the title above.

The worked solution is on the next slide — try it yourself first.

Integer → binary string

```
num = int(input('Enter a whole number: '))
if num < 0:
    isNeg = True
    num = abs(num)
else:
    isNeg = False
result = ''
if num == 0:
    result = '0'
while num > 0:
    result = str(num%2) + result
    num = num//2
if isNeg:
    result = '-' + result
print(result)
```

What about fractions?

- $3/8 = 0.375 = 3 \cdot 10^{-1} + 7 \cdot 10^{-2} + 5 \cdot 10^{-3}$
- Multiply by a power of 2 big enough to get a whole number, convert to binary, then divide by the same power of 2.
- $0.375 \cdot (2^3) = 3$ (decimal).
- Convert 3 to binary → 11.
- Divide by 2^3 (shift right) → 0.011 (binary).

Task

Decimal fraction → binary

Write (or trace) the Python program described by the title above.

The worked solution is on the next slide — try it yourself first.

Decimal fraction → binary

```
x = float(input('Enter a decimal number between 0 and 1: '))
p = 0
while ((2**p)*x)%1 != 0:
    print('Remainder = ' + str((2**p)*x - int((2**p)*x)))
    p += 1
num = int(x*(2**p))
result = ''
if num == 0:
    result = '0'
while num > 0:
    result = str(num%2) + result
    num = num//2
for i in range(p - len(result)):
    result = '0' + result
result = result[0:-p] + '.' + result[-p:]
print('The binary representation of the decimal ' + str(x) + ' is ' + result)
```

Newton-Raphson

- General approximation algorithm to find roots of a polynomial in one variable.
- **$p(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$**
- Want to find r such that p(r) = 0.
- For example, to find the square root of 24, find the root of $p(x) = x^2 - 24$.
- Newton showed that if g is an approximation to the root, then $g - p(g)/p'(g)$ is a better one (p' is the derivative of p).

Newton-Raphson — simple case

- Simple case: $cx^2 + k$.
- First derivative: $2cx$.
- So if the polynomial is $x^2 + k$, then the derivative is $2x$.
- **Given a guess g for the root, a better guess is: $g - (g^2 - k) / 2g$**
- This gives us another way of generating guesses, which we can check — very efficient.

Task

Newton-Raphson — square root

Write (or trace) the Python program described by the title above.

The worked solution is on the next slide — try it yourself first.

Newton-Raphson — square root

```
epsilon = 0.01
y = 24.0
guess = y/2.0
numGuesses = 0
while abs(guess*guess - y) >= epsilon:
    numGuesses += 1
    guess = guess - (((guess**2) - y)/(2*guess))
print('numGuesses = ' + str(numGuesses))
print('Square root of ' + str(y) + ' is about ' + str(guess))
```



NEW TOPIC

Functions

Reusable, named chunks of code

Functions

- Write reusable pieces / chunks of code, called functions.
- Functions are not run in a program until they are 'called' or 'invoked'.
- **Function characteristics:**
  - has a name
  - has parameters (0 or more)
  - has a docstring (optional but recommended)
  - has a body

Task

Defining and calling a function

Write (or trace) the Python program described by the title above.

The worked solution is on the next slide — try it yourself first.

Defining and calling a function

```
def is_even( i ):
    """
    Input: i, a positive int
    Returns True if i is even, otherwise False
    """
    print("hi")
    return i%2 == 0

is_even(3)
```

Default arguments

```
def func(name,last_name,reverse):
        if reverse==True:
                return last_name,
name
        else:
                return name,
last_name
print (func("Maksim", "Iavich",False))
```

```
def func(name,last_name):
        if reverse==True:
                return last_name,
name
        else:
                return last_name,
name
print (func("Maksim", "Iavich"))
```

Task 1

★ NEW

Hash a password with SHA-256

- **Never store passwords in plain text — store a one-way hash instead.**
- Write a function hash_password(pw) that returns the SHA-256 hex digest of the password.
- Show that changing a single character (even a space) produces a completely different hash.

Libraries / focus: *hashlib — cryptographic hashing*

Solution — SHA-256 hashing function

★ NEW

```
import hashlib

def hash_password(pw):
    """Return the SHA-256 hex digest of a password string."""
    data = pw.encode('utf-8')
    return hashlib.sha256(data).hexdigest()

print(hash_password('correcthorse'))
print(hash_password('correcthorse '))   # one space -> different hash
```

Task 2

★ NEW

Generate a QR code for a link

- Use the qrcode library to encode a URL into a QR image (e.g. a link to a security report).
- Save it as a PNG that can be printed or shared.
- **Install first: pip install qrcode[pil]**

Libraries / focus: *qrcode — turning data into scannable images*

Solution — writing a QR code

★ NEW

```
# pip install qrcode[pil]
import qrcode

url = 'https://cu.edu.ge'
img = qrcode.make(url)
img.save('link.png')
print('QR code saved to link.png ->', url)
```



END OF MEETING 2

Recap

Numbers in binary, roots by Newton-Raphson, and reusable functions